

From informative to insightful: redefining the measurement of integrated reporting quality

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Abstract

Purpose – As investors seek greater transparency in corporate strategy, prospects and sustainability performance, a robust measure of integrated reporting quality (IRQ) has become increasingly important. However, a lack of consensus on how IRQ should be assessed has prompted calls for refinement (De Villiers *et al.*, 2017). This study addresses the limitations of current “checklist”-based measures by developing an index to assess the depth and breadth of disclosures.

Design/methodology/approach – The revised measure integrates the guiding principles, content elements and fundamental concepts of the Integrated Reporting (IR) Framework into a single, intertwined construct. Its development involved consultations with experts at the Value Reporting Foundation (VRF) and was pilot-tested through the VRF’s gap analysis program. This study applies the measure to integrated reports from 51 S&P 1200 companies and compares them with traditional IRQ measures.

Findings – The revised measure identifies greater variation in IRQ than traditional approaches, distinguishing reports that are merely “informative” (i.e. referencing the IR Framework) from those that provide deeper insights into the organization’s value-creation story (“insightful”) and those that demonstrate coherence and reliability (“credible and connected”).

Practical implications – The measure offers researchers, standard-setters and practitioners a structured basis for evaluating integrated and sustainability-related reports more consistently across firms and periods, thereby improving report comparability and transparency.

Originality/value – This study advances IRQ measurement beyond checklist compliance by introducing an integrated, multidimensional tool that maps guiding principles onto content elements of the IR Framework to differentiate among levels of disclosure quality.

Keywords Broader corporate reporting, Integrated reporting, Integrated reporting framework, Integrated reporting quality, Integrated thinking

Paper type Research article

1. Introduction

Developments in corporate reporting over the past decade, especially the release of International Sustainability Standards Board (ISSB) Standards (IFRS S1 and IFRS S2) in 2023, have highlighted the growing importance of broader corporate reporting and stakeholders’ accountability. Most corporations, particularly listed entities, are expanding their annual reports beyond traditional financial statements to provide a more comprehensive perspective of corporate decision-making and value-creation processes. Modern annual reports are usually supplemented with additional disclosures, including those related to corporate governance and strategy, business model, stakeholder relationships and social and environmental (or sustainability) impacts. In response to increasing stakeholder demand for broader accountability, there is a clear trend toward integrating financial and non-financial information within an annual report (e.g. de Villiers *et al.*, 2017; Obeng *et al.*, 2021). This integration has also been publicly encouraged by the chairs of the International Accounting

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Standards Board (IASB) and ISSB to “drive high-quality corporate reporting and connectivity between financial statements and sustainability-related financial disclosures” (Dimes and de Villiers, 2024, p. 2).

While the adoption of IR Framework is not universal, many organizations, and a growing number of corporate regulators [e.g. the UK’s Companies Act 2006 (Strategic Report and Directors’ Report) Regulations 2013], address key elements of the IR Framework. An integrated report is defined as “a concise communication about how an organization’s strategy, governance, performance and prospects, in the context of its external environment, lead to the creation of value in the short, medium and long term” (IIRC, 2021, p. 10). KPMG (2022) finds that 55% of N100 companies in the Middle East and 30% in the Asia Pacific have adopted integrated reporting. Prior research from South Africa, where IR is mandated, finds capital market benefits associated with integrated reporting quality (e.g. Barth *et al.*, 2017; Zhou *et al.*, 2017). Proponents of integrated reporting argue that the integrated report is an ideal location for sustainability disclosures as the IR Framework emphasizes how non-financial information about *capitals*, business models and strategy come together to depict value creation (Songini *et al.*, 2023) [1].

Credible measurement of integrated reporting quality (IRQ) requires careful consideration of the distinctions between environmental, social and governance (ESG) reporting and integrated reporting (de Villiers *et al.*, 2017). ESG reporting has a broader stakeholder focus and involves reporting on an organization’s social and environmental impact. By contrast, an annual report prepared in accordance with the IR Framework is focused on the sources of enterprise value creation, competitive advantage and the sustainability of the organization and is primarily directed at investors and financiers, while being of interest to other stakeholders. Due to the different focuses and target audiences, an organization’s social or environmental impact will only appear in an integrated report if it is material to enterprise value creation.

Prior studies have employed various proxies for IRQ, including firm’s ESG performance (e.g. Melloni *et al.*, 2017; Obeng *et al.*, 2021; Serafeim, 2015) or checklist approaches based on content elements of the IR Framework (e.g. Zhou *et al.*, 2017). Building on prior research, the primary objective of this study is to develop a comprehensive IRQ measure that intertwines the content elements, guiding principles and fundamental concepts of the IR Framework. Rather than using the guiding principles of the IR Framework as checklist items (e.g. Liu *et al.*, 2019; Haji and Anifowose, 2016; Ruiz-Lozano and Tirado-Valencia, 2016; Stent and Dowler, 2015), our revised measure uses guiding principles to distinguish how organizations apply the IR Framework’s content elements in their reporting.

We validated the measure through consultations with experts at the Value Reporting Foundation (VRF), who applied the measure as part of the VRF’s gap analysis programs. The measure was accordingly updated following feedback from the VRF staff. Using a sample of 51 S&P 1200 companies that had prepared integrated reports in 2022, we compare our new measure to commonly used proxy measures of IRQ and find mostly non-significant correlations. The non-significant correlation with ESG performance measures (*Alt3_ESGScore* and *Alt4_ESGGrade*) is consistent with the argument in de Villiers *et al.* (2017) that proxy measures based on ESG performance do not provide meaningful insights into IRQ. The non-significant correlation with the dummy variable measuring IR adoption (*Alt5_ReferenceIR*) or the number of pages in the annual report (*Alt1_NoPages*) is perhaps not surprising, as all companies in the sample produce integrated reports and highlight deficiencies with these measures. The distribution of results suggests that the *DIRC_IRQ* discerns significantly greater variation in reporting quality than the traditional proxies, particularly by mitigating the “ceiling effect” often observed in simple checklists.

This paper makes several contributions. Following calls to refine the measurement of IRQ (e.g. de Villiers *et al.*, 2017), this study’s core contribution is the development of a more refined measure that blends guiding principles, content elements and fundamental concepts of the IR

Framework into a single evaluative construct. Unlike existing proxies, the proposed measure can be used to distinguish an “*informative*” report (i.e. merely referencing the IR Framework) from reports that provide insights into the organization and its value creation story (*insightful*), and those that include information that is “*credible and connected*”. The new measure accordingly responds to calls for a more direct measure of IRQ that goes beyond checklist compliance with the IR Framework (Pistoni *et al.*, 2018).

Understanding the underlying mapping technique that links guiding principles to each of the content elements and fundamental concepts of the IR Framework might support and deepen both preparers’ and users’ understanding of integrated reporting and integrated thinking. The measurement approach is especially beneficial in the preliminary stages of implementing integrated reporting in clarifying the connections among IR Framework’s components. Indeed, this benefit to preparers was confirmed by VRF staff during the pre-testing stage of the gap analysis project.

The methodology provides a template for embedding artificial intelligence (AI) in measuring IRQ across the reporting population. As reporting evolves and improves over time, the proposed measure might assist researchers in exploring the determinants or implications of IRQ. Our study may also be useful to the IFRS Foundation as it contemplates a conceptual framework for corporate reporting that builds on the IR Framework.

2. Literature review

2.1 Broader corporate reporting

With the increasing pressure on companies to focus on sustainability and broader accountability, a growing global movement calls for companies to disclose information outlining their sustainability performance (Christensen *et al.*, 2021; de Villiers *et al.*, 2023; Dhaliwal *et al.*, 2011). Sustainability reporting has been largely voluntary across most international jurisdictions, with disclosures found in either the annual reports or standalone reports, often labelled as ESG, CSR or Sustainability Reports. Companies have typically presented sustainability information with reference to a range of voluntary reporting frameworks [e.g. Global Reporting Initiative (GRI), United Nations Sustainable Development Goals (SDGs), Task Force on Climate-Related Financial Disclosures (TCFD), Sustainability Accounting Standards Board (SASB) standards and the Climate Disclosure Project (CDP)]. Christensen *et al.* (2021) observed that investors frequently complained about a lack of comparable and verifiable information. In response, the ISSB took a significant step toward consolidating and standardizing sustainability reporting in 2023 by issuing its inaugural sustainability disclosure standards (IFRS S1 and IFRS S2), which consolidated content from existing frameworks, including TCFD, CDP, the IR Framework and the SASB standards. The stated purpose of the ISSB standards is to deliver a global baseline of sustainability disclosures to inform capital markets [2].

Generally, two broad user focuses have emerged within regulatory initiatives and standard-setting processes regarding sustainability reporting, as illustrated in Figure 1.

Under popular voluntary frameworks, like GRI or SDGs, sustainability reports target a wider group of stakeholders and foster transparency about an organization’s impact on the environment and society. This approach has emerged with the concept of double materiality, which emphasizes the interdependence between the company and its socio-environmental context, where sustainability reporting influences corporate behaviour by encouraging desirable and discouraging undesirable activities (Christensen *et al.*, 2018). The concept of double materiality has been embedded into the Corporate Sustainability Reporting Directive (CSRD) in Europe, which mandates companies to report on both their financial (i.e. how sustainability issues affect enterprise value) and impact materiality (i.e. how the company impacts the environment and society). The CSRD broadens reporting requirements and introduces the European Sustainability Reporting Standards (ESRS), ensuring greater transparency, comparability and accountability in sustainability disclosures among European

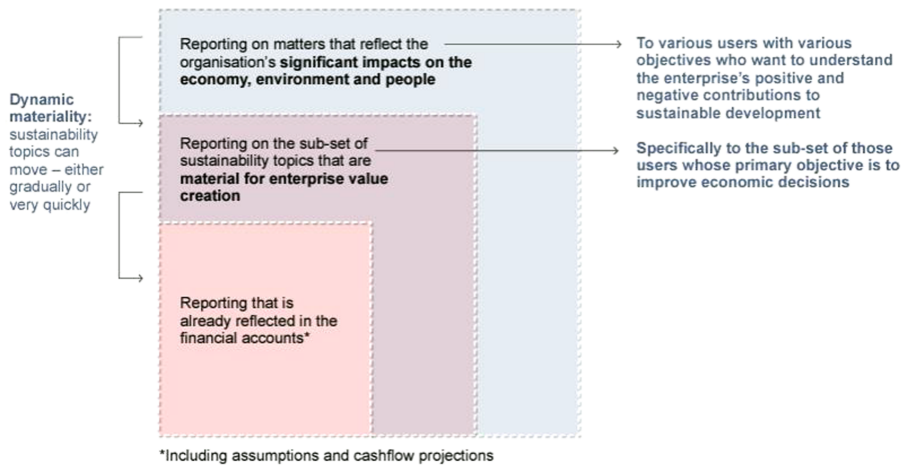


Figure 1. Dynamic materiality diagram. Alliance of Sustainability Reporting Organizations (2020)

companies. While information related to socio-environmental risks and rewards is valuable for fostering sustainable practices, it may not necessarily translate into financial outcomes. Therefore, such information may not be material to investors, and it is labelled as “non-financial sustainability information”.

The second approach, aligned with the ISSB standards, focuses on sustainability-related financial information that affects companies’ prospects. This investor-centric approach recognizes that sustainability information can evolve over time and impact companies’ prospects (ISSB, 2022; Kuh *et al.*, 2020). The IR Framework, aligned with this approach, emphasizes that only sustainability information material to enterprise value creation should be included in an integrated report. This makes the integrated report an ideal place for sustainability-related disclosures, as it integrates material sustainability-related financial information into the company’s strategic context and value creation processes and demonstrates how sustainability issues affect long-term business performance.

The IFRS Foundation’s acquisition of the IR Framework in 2022 marked an important milestone in unifying the corporate reporting system. While the role of the IR Framework in the future of corporate reporting remains under discussion, a joint statement by the Chairs of the International ISSB and the IASB in May 2022 endorsed the future of integrated reporting and specifically the IR Framework and Integrated Thinking Principles: “*We are convinced that the Integrated Reporting Framework drives high-quality corporate reporting and connectivity between financial statements and sustainability-related financial disclosures which improves the quality of information provided to investors. Therefore, we strongly encourage continued use of the Integrated Reporting Framework and the Integrated Thinking Principles underpinning it*” (IFRS Foundation, 2022a; Dimes and de Villiers, 2024, p. 2). This statement highlights ongoing efforts to improve connectivity between financial and sustainability-related financial information, aligning with the goal of providing relevant, high-quality information to assist investors and other stakeholders in their decision-making process.

2.2 Strands of IRQ measurement

Research investigating IRQ has adopted various proxies and assessment methods, including binary variables, ordinal scoring systems, award-based assessments and a variety of textual

analysis techniques [3]. While there is no consensus on the best approach, the literature can be categorized into three distinct strands: (1) Empirical academic research; (2) Practitioner-oriented and awards; and (3) Indirect Proxies. Each strand reflects different levels of objectivity and adherence to the IR Framework.

2.2.1 Empirical academic strand. The empirical academic strand plays a critical role by developing replicable instruments to test the IR Frameworks' components across large populations. Within this strand, measures are developed based on the conceptual theory and into verifiable evidence, which is mostly gathered through the content analysis method. The simplest method is using a binary variable (Yes/No) to assess whether a company has prepared an integrated report. Prior research classified a company as preparing an integrated report if it publishes a report labelled "Integrated Report" or if the company is listed on the IIRC "Examples Database" (Gal and Akisik, 2020; Garcia-Sanchez and Noguera-Gomez, 2017; Hsiao et al., 2021; Landau et al., 2020; Mervelskemper and Streit, 2017; Muttakin et al., 2020; Vena et al., 2020; Wahl et al., 2020). The problem is that this approach ignores whether the integrated report has been prepared in accordance with the IR Framework.

This concern is addressed in studies analysing IRQ through manual content analysis of annual reports. A checklist, typically comprising *content elements*, *guiding principles* and/or *fundamental concepts* of the IR Framework, is used to score annual reports on whether the abovementioned IR Framework elements have been covered in the report. Following this approach, Stent and Dowler (2015) applied a report checklist based on the IIRC Prototype Framework to evaluate the 2011 annual reports of four New Zealand companies nominated as best practice reporters by the New Zealand Institute of Chartered Accountants. The scoring system comprised 33 predominantly binary questions [4]. Similarly, Marx and Mohammadali-Haji (2014) assessed disclosures in integrated reports, annual reports and websites of the top 40 listed companies in South Africa (FTSE/JSE Top 40) using 18 items (predominantly content elements) from the IIRC prototype framework coded on binary scales (Yes/No).

Other studies applied binary scoring methods across broader samples and contexts. Kılıç and Kuzey (2018) assessed the annual and sustainability reports of the 100 largest companies listed on the Borsa Istanbul 100 Index using a checklist of 50 items grouped into seven categories corresponding to the seven content elements in the IIRC prototype Framework (2011). Islam (2020) applied a reporting checklist to the annual reports of 20 companies listed on the Bangladesh Securities and Exchange Commission (BSEC), comprising 36 items across eight categories corresponding to the eight content elements of the IR Framework. Tiron-Tudor et al. (2020) analysed 98 integrated reports from 61 European companies published on the IIRC website (Examples Database) from 2013 to 2017 by using a scoring system of 28 binary questions weighted equally across the eight content elements of the IR Framework. Cooray et al. (2020a) extended this approach to the integrated reports of 39 companies listed on the Colombo Stock Exchange (CSE) by applying a checklist of 38 disclosure items classified under the eight *content elements* of the IR Framework using predominantly binary scoring resulting in a maximum total score of 74. [5] While the binary content analysis offers simplicity and objectivity, it primarily captures compliance with the IR Framework rather than the depth or richness of related disclosures.

To address the limitations of binary scoring, some studies have adopted an ordinal scaling method. Lee and Yeo (2016) assessed 822 integrated reports from the Johannesburg Stock Exchange (JSE) between 2010 and 2013. The scoring system comprises five questions for each of the eight *content elements* from the IR Framework; these 40 questions are rated on a 5-point ordinal scale from 0 (non-compliance) to 5 (strong compliance). Similarly, Haji and Anifowose (2016) assessed 82 integrated reports from the top 100 listed companies on the JSE during 2011–2013 ($n = 246$). The scoring comprised 52 items graded on a 4-point scale [6] within eight categories that correspond to the guiding principles from the IIRC prototype framework and the additional category, "*risk management practices*". Ruiz-Lozano and Tirado-Valencia (2016) applied an ordinal scoring system to 21 international industrial companies on the IIRC Integrated Reporter database in 2013/2014.

Their rating comprised 32 items within seven categories, corresponding to the seven guiding principles from the IR Framework [7]. Zhou *et al.* (2017) assessed the integrated reports of 132 South African companies listed on the JSE from 2009 to 2012. The coding comprises 31 questions, scored between zero and one, across seven categories that align with content elements in the IIRC Prototype Framework and an additional category, “other”, which includes questions on conciseness, materiality and board sign-off [8]. Liu *et al.* (2019) developed a unique reporting checklist that was applied to the annual reports of 5 Australian companies [9]. The measure comprises 22 categories corresponding with the *content elements*, *guiding principles* and *capitals* from the IR Framework. The categories are further divided into *second-level subcategories* following GRI 2013 Guidelines, resulting in a total of 941 disclosure items [10].

Pistoni *et al.* (2018) developed a more sophisticated scoring model comprising four categories measured using 23 scoring items. The four disclosure categories are *background* (covering issues like motivation for adopting integrated reporting, beneficiaries of the report, CEO commitment), *assurance and reliability* (whether the report undergone internal audit, third-party verification or has received an award), *content* (whether the report includes the eight content elements and two (of the three) fundamental concepts of the IR Framework) and *form* (whether the report is readable and clear, concise and accessible). Pistoni *et al.* (2018) use a combination of binary (i.e. for the background and assurance and reliability categories) and ordinal scales (for the content and form categories). [11] Several studies have subsequently applied this IRQ measure (e.g. Qaderi *et al.*, 2021; Raimo *et al.*, 2021a, b; Songini *et al.*, 2021; Vitolla *et al.*, 2019a, b, c, 2020a, b).

More recently, studies have focused on using machine learning to measure aspects of the IR Framework. For example, Caglio *et al.* (2020) apply textual analysis to the integrated reports of approximately 80 listed firms on the JSE during the period 2011 to 2016 to identify key textual characteristics underpinning IR disclosure quality (readability, conciseness and credibility). The measures focus on the amount (length) and style (readability and tone) of disclosure in the integrated report, and the presence of a third-party assurance statement, thus addressing three of the seven guiding principles of the IR Framework. Similarly, Quarchioni *et al.* (2021) use machine learning algorithms to identify key vocabulary clustered around the six *capitals* and utilize network text analysis to investigate if the identified *capitals* are connected within the integrated report. However, the measure disregards other content elements of the IR Framework. Melloni *et al.* (2017) use a combination of textual analysis of firms’ IR and ESG disclosure scores to explore the balance between two guiding principles of the IR Framework, *conciseness* and *completeness* [12].

The recent evolution of IRQ measurement is best exemplified by Barth *et al.* (2025), who developed a sophisticated automated measure using a machine-based textual analysis along four dimensions: “*textual attributes, topical content, capitals, and financial versus sustainability information – to measure different attributes of IRQ*” (p. 22). This approach represents an advancement over prior textual analysis by training a Large Language Model (LLM) to recognize specific IR content elements across thousands of reports with a high degree of precision. While sophisticated machine learning models resolve the scalability limitations of manual coding and improve the objectivity of “presence-based” scoring, their capacity to assess principles such as connectivity and “integrated thinking” behind those disclosures remains limited. Manual indices such as those developed in the present study are likely to be essential for providing the qualitative nuances for “inputs” required to train and refine models like those implemented in Barth *et al.* (2025).

2.2.2 Practitioner-oriented and award-based strand. In contrast to the empirical academic strand, practitioner-oriented and award-based measures serve a distinct role by establishing the industry’s best practices. These measures prioritize expert judgment to capture the nuances of “excellence” that automated checklists often miss. The EY (South Africa) Ranking (2012–2022) is one of the earliest and most frequently cited measures of IRQ (e.g. Barth *et al.*, 2017; Mans-Kemp and Van der Lugt, 2020; Maroun, 2018; Omran *et al.*, 2021; Wang *et al.*, 2020).

The measure comprises 15 items (7 *guiding principles* and 8 *content elements*) from the IR Framework (IIRC, 2021) rated on a 10-point scale and assessed by three independent experts whose aggregate scores rank companies into four categories: “Excellent”, “Good”, “Average” and “Progress to be made”. [13] While EY does not publish a detailed grading rubric, the process described in the annual EY reports involves discussion and conferring among three experts, suggesting that the measure is more than a checklist collation of compliance with the IR Framework [14].

The Institute of Certified Management Accountants (CMA) of Sri Lanka also conducts annual “CMA Excellence in Integrated Reporting” awards. Companies voluntarily submit their reports for assessment, which is based on 17 criteria, including 7 *guiding principles*, 8 *content elements* and 2 *fundamental concepts* (i.e. value creation and the *capitals*) on a 10-point scale basis, except for the “Strategic Focus and Future Orientation” element with 20 points and “Value Creation” with 30 points. A panel of local and international experts compares scores and, following discussion, makes the final judgment on the winning companies. While the checklist-based review process measures adherence to the IR Framework, the panel discussion likely ensures that the assessment of IRQ goes beyond a checklist.

2.2.3 Indirect proxies strand. A final, distinct strand of literature utilizes indirect proxies to facilitate large-sample capital market research, providing a cost-effective way to observe reporting outcomes over time. Measures of a firm’s ESG performance are frequently used as proxies for IRQ. For example, Melloni *et al.* (2017) used Bloomberg’s ESG disclosure score to proxy the “completeness” of an integrated report. The most commonly used proxy is the Thomson Reuters ASSET4 ESG performance index (e.g. Obeng *et al.*, 2021; Serafeim, 2015). Serafeim (2015, p. 38) argues that the proxy “. . . reflects a company’s capacity to convincingly show and communicate that it integrates the economic (financial), social and environmental dimensions into its day-to-day decision-making processes.” However, the ASSET4 ESG score primarily reflects ESG outcomes (organization’s impact on society and the environment) and does not directly measure the level of connectivity within the annual report, or how a firm’s strategy for using *capital* in its business model creates long-term value, which is indicative of better IRQ. De Villiers *et al.* (2017) accordingly question the “construct validity” of ASSET4 as a proxy for integrated reporting. As such, ASSET4 is, at best, an imperfect proxy measure of IRQ.

3. Proposed measure

The content elements, fundamental concepts and guiding principles of the IR Framework represent distinct yet interconnected dimensions of reporting quality. Content elements specify the information to be disclosed, while the guiding principles and fundamental concepts assist in evaluating the quality of content element disclosure. This study integrates guiding principles and fundamental concepts into a five-point grading rubric to assess the disclosure quality of content elements.

3.1 Components of the measure

Our measure consists of 10 grading items: (1) External Environment and Organization overview; (2) Governance; (3) Business Model; (4) Risks and Opportunities; (5) Strategy and Resource Allocation; (6) Performance; (7) Outlook; (8) Basis of Presentation and Preparation; (9) The *Capitals*; and (10) Stakeholder Relationships [15]. The first eight items correspond directly to the content elements in the IR Framework. We add two grading items: (1) Given the prominence of the *capitals* in explaining a firm’s value creation story and their identifiable nature, we include “the *capitals*” as a standalone grading item; and (2) “Stakeholder Relationships” is included as a standalone grading item as managing key stakeholder relationships is critical to a firm’s value creation process (Argandona, 2011).

The selection and equal weighting of these ten items are grounded in the fundamental principle that integrated reporting quality is an “intertwined construct” rather than a collection of disparate disclosure qualities. While some professional awards assign higher weights to specific elements, such as “Strategic Focus” (e.g. the CMA of Sri Lanka), we have opted for an equal-weighting schema because the IR Framework emphasizes the “connectivity of information” as its central guiding principle. This approach reflects the theoretical requirement for a balanced and holistic narrative across the entire value-creation story; for instance, an insightful discussion on “Risks” is of limited utility to an investor if it is not explicitly linked to “Strategy” or “Performance”. Furthermore, maintaining a consistent weight across all 10 items minimizes researcher bias and ensures that the *DIRC_IRQ* remains a standardized and comparable benchmark across diverse global industries.

We use a five-point grading scale to evaluate the ten grading items. We incorporate five of the seven guiding principles from the IR Framework into our five-point grading rubric to discern a report’s quality, including: (1) Strategic focus and future orientation; (2) Connectivity of information; (3) Conciseness; (4) Reliability and Completeness; and (5) Consistency and Comparability. Regarding the remaining 2 guiding principles, the inclusion of “Stakeholder Relationships” as a grading item was previously discussed, and finally “Materiality” is included within the content element “Basis of preparation and presentation”, which specifies disclosure of the organization’s materiality determination process [16]. Further, we use the presentation of performance and outlook data in our five-point grading scale across all ten content elements. For example, we use the presentation of key performance indicators (KPIs) to assist in grading “Governance”, “Business Model”, “Strategy and Resource Allocation”, “Risks and Opportunities”, “The *Capitals*” and “Stakeholder Relationships”.

3.2 Grading scale

Each of the ten grading items is evaluated on a five-point ordinal scale: Foundational (score 1 out of 5); Developing (2); Informative (3); Insightful (4); and Connected and Credible (5). The rubric’s five-point scale promotes consistent and heterogeneous report ratings, unlike prior studies, where the basis for rating is often unclear and treated as a “black box” (e.g. EY South African Measure, 2012–2022; Zhou *et al.*, 2017). A summary of the marking rubric is presented in Table 1.

3.3 Validating the design of the *DIRC_IRQ* measure

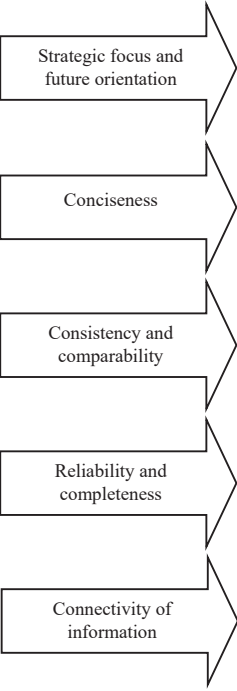
The proposed measure (*DIRC_IRQ*) was validated through the VRF 2021 Gap Analysis Program. To ensure high construct validity, the measure was developed through a three-stage feedback loop involving VRF representatives: (1) An initial review was conducted to align the 10 items with the IR Framework with three representatives from VRF, two representatives from the industry (i.e. a Big-4 company) and four representatives from two universities; (2) Pilot testing was done on 10 international reports which identified that “connectivity” required more granular indicators; and (3) Final refinement, where performance and outlook data were integrated directly into the grading scale across all elements. Following this refinement, the measure was then used to evaluate the quality of 25 international integrated reports voluntarily submitted for critique as part of the program, for which each company received detailed feedback outlining the strengths and weaknesses of their reporting based on the scoring rubric. Thus, the *DIRC_IRQ* measure was validated through application and demonstrated its construct validity.

4. Data and methods

4.1 Sample selection and description

The IFRS Foundation identified 185 S&P Global 1200 companies that prepared integrated reports in 2022 [17] and [18]. Among those 185 entities, there were 3 regions: 23 in the

Table 1. Grading scale of the measure

IR Framework Guiding Principles		Grade	Criteria
	Strategic focus and future orientation	1 Foundational	Information in the report is presented in silos, and is primarily focused on financial information.
		2 Developing	‘Boilerplate’ reporting with some discussion on the content elements. Strategy is described, but lacks discussion on <i>capitals</i> used to achieve strategic goals. Information is presented in silos and lacks logical flow.
	Conciseness	3 Informative	Informative presentation of content elements, with credible explanations of strategy, <i>capitals</i> , and insights on value creation. However, limited insightful analysis, particularly on strategy implementation and the <i>capitals</i> used in the business model. Connectivity among content elements is lacking, and its reliability is difficult to assess.
	Consistency and comparability	4 Insightful	The report aligns with the IR Framework and offers insight into the business, performance, and prospects. An insightful report explains ‘what’ the strategy is, ‘with’ which <i>capitals</i> the strategic goals can be achieved, ‘how’ value is created through the business model, and ‘when’ the strategic objectives are expected to be achieved. The report includes some analyses (e.g., ratio, sensitivity, scenario, or qualitative analysis), and connectivity among discussions, diagrams, and metrics is evident, reflecting integrated thinking. Limited independent assurance is introduced.
	Reliability and completeness	5 Credible and Connected	The report presents interrelated content elements, enabling users to navigate concise, connected information and understand how the business creates value. The report has an easy-to-navigate structure, ensuring transparency and reliability. It presents a complete picture of the business, including its performance, prospects, and competitive edge, in both positive and negative angles. The board acknowledges its accountability for the integrity of the report, as evidenced by integrated report assurance.
	Connectivity of information		

Americas, 87 in Asia-Oceania and 75 in Europe. Following a simplified sample selection method [19], the estimated ideal sample size for 185 IR, with a 90% confidence level and 10% margin of error, would be 50 reports. We use the Python random selection model to select 17 entities from each region, resulting in 51 entities in total.

4.2 Method

To assess the effectiveness of our proposed measure, we compare its results with those of the five most commonly used proxies of IRQ for which data are publicly available [20].

- (1) *Alt1_NoPages* is a simple measure based on the number of pages in the annual report, excluding the financial statements, divided by the total length of the annual report

(adapted from the approach in [Melloni et al., 2017](#); [Caglio et al., 2020](#)). This measure makes the simplifying assumption that more discussion reflects more information relevant to stakeholder understanding of the drivers of business prosperity.

- (2) *Alt2_ContentElement* captures whether a report referencing the *content elements* in the IR Framework (e.g. [Cooray et al., 2020](#); [Haji and Anifowose, 2016](#); [Islam, 2020](#); [Kılıç and Kuzey, 2018](#); [Marx and Mohammadali-Haji, 2014](#); [Stent and Dowler, 2015](#); [Tiron-Tudor et al., 2020](#)). This measure is usually constructed as a continuous variable between 0 (no content elements covered) and 8 (all content elements covered)
- (3) Refinitiv’s ESG score (i.e. previously Thomson Reuters ASSET4) ([Obeng et al., 2021](#); [Serafeim, 2015](#)) measures a company’s relative ESG performance, commitment and effectiveness across 10 categories, weighted by industry, with the final score in the 0–100% range (hereafter, *Alt3_ESGScore*). In addition, Refinitiv has created a scale to grade ESG performance, ranging from A+ (the best performance) to D- (worst performance). We transformed the scale from 1 (grade D-) to 12 (grade A+) and labelled it *Alt4_ESGGrade*.
- (4) *Alt5_ReferenceIR* is a dummy variable that captures whether the annual report refers to the IR Framework (e.g. [Garcia-Sanchez and Noguera-Gamez, 2017](#); [Gal and Akisik, 2020](#); [Hsiao et al., 2021](#); [Landau et al., 2020](#); [Mervelskemper and Streit, 2017](#); [Muttakin et al., 2020](#); [Vena et al., 2020](#); [Wahl et al., 2020](#)).

For our sample of 51 companies, two researchers independently coded *DIRC_IRQ*, *Alt1_NoPages*, *Alt2_ContentElement* and *Alt5_ReferenceIR*. Scores were compared, and when differences occurred, a third researcher independently rated the report. The final score was determined following a discussion between all researchers. The other two measures, *Alt3_ESGScore* and *Alt4_ESGGrade*, are drawn from the Refinitiv ESG database.

5. Empirical results

5.1 Descriptive statistics and correlation matrix

[Table 2](#) provides descriptive statistics for *DIRC_IRQ* and alternative measures of IRQ for the 51 global integrated reporters. The median value for *DIRC_IRQ* is 30 of the maximum value of 50, indicating an average (*informative*) adherence to the IR Framework. We observe a near-perfect score (7.75 of 8) for *Alt3_ContentElement* and 100% for *Alt5_ReferenceIR*, which is expected as the 51 entities prepare integrated reports, highlighting the “ceiling effect” in checklist-based measures. The mean value of *Alt1_NoPages* is 0.73, indicating that, on average, 73% of the annual report contains information in addition to the statutory financial statements. Finally, we observe high ESG scores with a median value of 80% for *Alt4_ESGScore* and a median score of 10 of 12 (83%) for *Alt4_ESGGrade*. The distribution of

Table 2. Descriptive statistics

Medium	Mean	Median	Minimum maximum	Standard deviation	Potential range
<i>DIRC_IRQ</i>	30.33	30	19–43	5.35	0–50
<i>No of Pages</i>	51.35	47	6–210	28.17	0–210
<i>Alt1_NoPages</i>	0.73	0.66	0.25–1	0.26	0–1
<i>Alt2_ContentElement</i>	7.75	8	5–8	0.69	0–8
<i>Alt3_ESGScore</i>	0.78	0.80	0.56–0.94	0.09	0–1
<i>Alt4_ESGGrade</i>	9.82	10	7–12	1.14	1–12
<i>Alt5_ReferenceIR</i>	1	1	1–1	0	0–1

results across the six measures suggests the *DIRC_IRQ* discerns greater variation in reporting quality than traditional proxies.

The distribution of values for the 51 global integrated reports across the ten grading items that comprise the *DIRC_IRQ* measure of integrated reporting quality is presented in Figure 2. The mode of distribution for 8 of the 10 grading items is a score of 3 (informative), indicating that most organizations provide an informative business description around these 8 elements. A mode of 2 (“developing”) is observed Governance because most companies did not disclose how those charged with governance support the value-creation process. A mode of 4 (insightful) was observed for “Organizational overview and external environment” as most companies make an insightful disclosure about what the organization does and the circumstances under which it operates. No entity scored 5 for any of the 10 grading items.

We assess the efficacy of the alternative proxies of IRQ by assessing the correlation among measures. The Pearson correlation results for the 51 global integrated reporters are presented in Table 3. Other than *Alt2_ContentElement* (0.56), no significant correlation was observed between the *DIRC_IRQ* and the other IRQ proxies, which indicates that our new measure provided unique insight into IRQ [21]. The significant correlation with *Alt2_ContentElement* is expected as both measures are grounded in the IR Framework’s content elements. However, the lack of a “ceiling” in our proposed measure (standard deviation = 5.35) ensures it remains a more robust tool for identifying “insightful” and “credible” reporting.

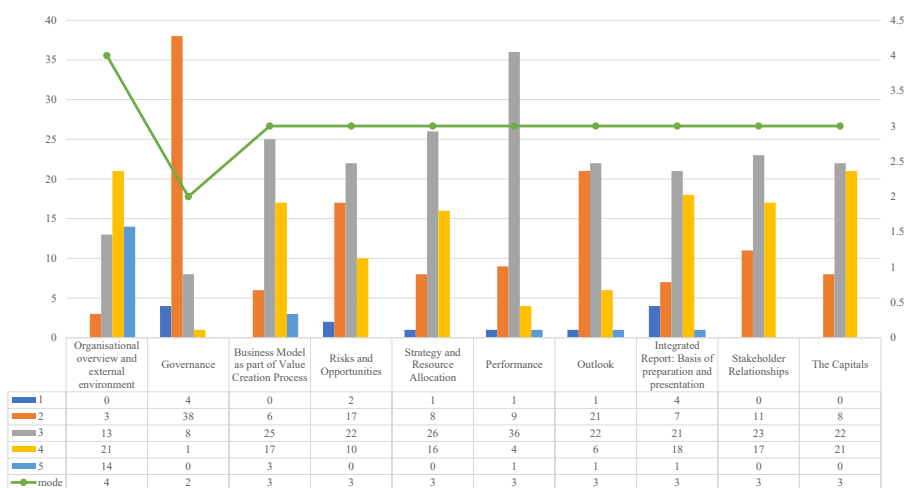


Figure 2. Score distributions

Table 3. Correlation matrix

	(1)	(2)	(3)	(4)	(5)
(1) <i>DIRC_IRQ</i>	1				
(2) <i>Alt1_NoPages</i>	0.22	1			
(3) <i>Alt2_ContentElement</i>	0.56***	−0.09	1		
(4) <i>Alt3_ESGScore</i>	0.09	−0.03	−0.03	1	
(5) <i>Alt4_ESGGrade</i>	0.05	−0.05	−0.02	0.96***	1

Note(s): *, **, *** indicates significance at 10%, 5% and 1% level

6. Conclusion

The transition toward a unified global baseline for sustainability-related financial disclosures, led by the IFRS Foundation and the ISSB, has re-emphasized the critical need for high-quality integrated reporting. This study addresses the long-standing challenge of measuring IRQ by developing a new measure that integrates content elements, guiding principles and fundamental concepts of the IR Framework. We validated and refined the measure through consultation with experts at the VRF and its application through VRF's Gap Analysis program in 2021. This study compares results from the proposed measure with alternative proxies of IRQ using the integrated reports issued by 51 S&P 1200 companies for the year 2022. Results suggest that *DIRC_IRQ* is more effective at discerning variation in IRQ than measures used in prior research, particularly for better quality reporters. Specifically, the *DIRC_IRQ* mitigates the "ceiling effect" found in checklist-based measures, which fail to differentiate quality once basic compliance is met. Further, findings confirm that proxies of ESG performance fail to effectively capture IRQ, consistent with [de Villiers et al. \(2017\)](#). This suggests that reporting quality is a distinct construct from sustainability performance, requiring specialized tools rather than general ESG scores.

We contribute to the literature by developing a comprehensive, nuanced measure of IRQ that serves as a reliable benchmark for emerging machine learning models. Our *DIRC_IRQ* measure complements automated advances by focusing on "depth" rather than just "breadth". For example, while an LLM can identify the existence of a "Business Model" or "Strategy" section, its capacity to reliably grade the coherence, synergy, or connectivity between these sections will be limited in the absence of training based on coding by experts. Thus, manual indices such as the *DIRC_IRQ* measure developed in the present study are essential for providing the "inputs" required to train and refine LLM models, providing a reliability benchmark that captures the qualitative nuances that current algorithmic models may conflate.

Further, reliable measurement of IRQ is important because it allows preparers and other users (e.g. policymakers and regulators) to evaluate the purported costs and benefits of integrated reporting. Prior research has also found evidence of positive economic consequences associated with high IRQ ([Barth et al., 2017](#); [Zhou et al., 2017](#)). Beyond market outcomes, our measure provides a structured basis to address the "integrated thinking" debate by helping researchers determine if high-quality reporting reflects substantive internal change or merely compliance. Future research might also explore whether greater integrated reporting quality demonstrates superior internal resource allocation or reduced information asymmetry in capital markets.

A mitigating factor against finding an IRQ associated with the market consequence (i.e. share price reaction) is the pre-existing ability of market experts and/or skillful analysts to access information on a firm's strategy and prospects. In other words, integrated reporting might not provide new information to skilled market analysts, so subsequent market consequences might not be observed. Nonetheless, future research might consider applying the *DIRC_IRQ* to longitudinal panel data to track how "connectivity" evolves under the IFRS S1 and S2 standards.

Table A1. Grading Matrix for *DIRC_IRQ*

Grading items	1 poor	2 developing	3 designated, identifiable and informative	4 concise and insightful	5 credible, connected and intuitive
Basis of Preparation and Presentation (Including Materiality)	The report does not explain how the organization determines what matters to include in the report or how those matters are quantified or evaluated	While the report may claim adherence to the Integrated Reporting (IR) Framework or its principles, it lacks substantive details on how material topics are identified, assessed, or incorporated into the report	The report explains how the organization determines which matters to include and how they are quantified or evaluated. It references key reporting frameworks or standards (e.g. the IR Framework, ISSB, SASB, or GRI) and outlines the materiality determination process. Additionally, it provides a summary of key judgments made in selecting material matters	The report contains a dedicated materiality section that explains how material matters are identified, ranked, and assessed. The sources of information used in this process are clearly specified. Additionally, the report enhances credibility by describing assurance mechanisms, such as director responsibility statements/limited external assurance, providing stakeholders with greater confidence in the integrity of the report	The report integrates the materiality determination process seamlessly within the report, linking material matters to relevant risks, opportunities, <i>capitals</i> , business model and stakeholder considerations. The report also ensures strong connectivity across disclosures by referencing related reports within the corporate reporting suite. Where limitations exist, such as restricted access to competitor data, these constraints are transparently acknowledged. Further enhancing credibility, robust assurance mechanisms, including reasonable assurance where applicable, are in place

(continued)

Table A1. Continued

Grading items	1 poor	2 developing	3 designated, identifiable and informative	4 concise and insightful	5 credible, connected and intuitive
Organizational Overview and External Environment	The report contains limited fundamental information about the organization's activities, structure, or market context, leaving stakeholders with limited understanding of the organization's operating landscape	The report includes a brief discussion of the organization's principal activities, purpose and mission. However, there is minimal or no reference to the external environment, such as market conditions, industry trends, or key external factors influencing the business	The organizational overview provides clear details on the organization's principal activities, structure (e.g. number of employees), purpose, mission, values and market position. The external environment is discussed with some reference to the organization's competitive landscape, including major suppliers, customers, or industry positioning	The report discusses the organization's purpose, culture, ethics and values, providing insights into how these aspects are evolving over time. The external environment discussion includes detailed narratives and quantitative information on industry trends, competitive positioning and key external factors such as technology shifts, demographic changes, climate- or nature-related risks. The discussion is supported by credible, externally sourced data where relevant	The report links the organization's purpose, culture, ethics and values with governance, strategy, risks, opportunities, business model and material matters. The external environment discussion highlights significant changes over time and considers the implications of key external factors (e.g. technology, demographic or climate change) on the business. This includes data sourced from credible external references, demonstrated through visuals such as graphics, hyperlinks or interactive elements. The discussion connects to strategic risks and opportunities, showing how external factors shape the company's future direction. Assurance on key external data sources, competitor benchmarking accuracy and macroeconomic assumptions used in forecasting

(continued)

Table A1. Continued

Grading items	1 poor	2 developing	3 designated, identifiable and informative	4 concise and insightful	5 credible, connected and intuitive
Stakeholder Relationship Management	There is no evidence that the organization considers the interests or legitimate needs of key stakeholders beyond shareholders	There is limited discussion on stakeholder engagement practices or how the organization understands and responds to stakeholder needs	The report provides an informative discussion on key stakeholder relationships, explaining why they are important for value creation. The report outlines the stakeholder engagement process and provides some evidence of engagement activities, such as the frequency of interactions with stakeholders (internal and external). There is an indication that the organization considers the legitimate needs and interests of all key stakeholders	The report not only describes key stakeholders and their importance to value creation but also explains how the organization manages these relationships. The report includes quantitative and qualitative KPIs, such as stakeholder engagement metrics (e.g. employee turnover, customer satisfaction scores, proportion of staff engaged in community service, innovation). It also provides a comparison of stakeholder engagement trends over time, evaluating the effectiveness of stakeholder relationship management in the current year compared to previous periods	The report demonstrates how stakeholder engagement impacts strategy, the business model and value creation. It provides a detailed evaluation of the effectiveness of stakeholder management, using SMART KPIs. The discussion links stakeholder priorities to business strategy, risk management, and long-term value creation. The report compares stakeholder engagement trends over time and may set future engagement goals, demonstrating a forward-looking approach to stakeholder relationship management. Validation of stakeholder mapping methodology, engagement outcomes and impact measurement, ensuring stakeholder interests are fairly represented

(continued)

Table A1. Continued

Grading items	1 poor	2 developing	3 designated, identifiable and informative	4 concise and insightful	5 credible, connected and intuitive
Governance	The report does not reference any applicable governance framework. There is no corporate governance statement or summary, and it is unclear whether the organization adheres to any national corporate governance code (e.g. ASX Corporate Governance Principles and Recommendations in Australia)	The report acknowledges adherence to a governance framework, but it may be boilerplate or checklist-based, focusing only on adherence to national governance principles without meaningful discussion. It may include a few key governance metrics, such as board diversity, skills matrix, or board experience, but does not elaborate on governance effectiveness	The report provides a factual summary of key areas of board focus during the year, highlighting how those charged with governance (TCWG) support value creation. There is some discussion of how the board monitors business performance, such as contributing to or reviewing the strategy, business model, or remuneration policies. The report may include basic performance measures of governance (e.g. code of conduct breaches, ethical breaches, whistleblowing reports, fraud incidents, or regulatory compliance issues), but minimal discussion on remedial actions	The report provides insights on the board's key areas of focus and their role in supporting value creation. It must include a responsibility statement from the board, acknowledging accountability for value creation and the integrity of the report. The board's oversight role is explained, with reports on the outcome from its engagement in strategy review, mentoring, ethical culture, talent management and emerging risks. The report provides quantitative governance performance indicators, including their role in strategic alignment, executive engagement and compliance and describes how breaches, governance failures or successes have been remediated. Additionally, it highlights governance's role in responding to emerging risks, such as pandemics, cybersecurity threats or opportunities, or climate-related governance issues	The report demonstrates active governance by the board. It includes a formal board responsibility statement, reinforcing the board's accountability for the report's integrity. The board's oversight role is clearly linked to strategy, risk management, business model, stakeholder engagement and innovation. SMART KPIs are used to assess board's contribution to value creation, its monitoring role, its involvement in strategic alignment, executive engagement and compliance and how breaches have been remediated (e.g. percentage of directors achieving personal governance KPIs, trend analysis of compliance incidents, or board effectiveness scores). It also explores future governance priorities, such as working alongside management to think about scenario planning for potential "black swan" events. Additionally, it provides evidence of independent board performance assessments and describes how governance structures evolve to address future challenges. Independent board assessment, validation of ethical compliance reporting and remuneration alignment with long-term value creation

(continued)

Table A1. Continued

Grading items	1 poor	2 developing	3 designated, identifiable and informative	4 concise and insightful	5 credible, connected and intuitive
Strategy and Resource Allocation	The report provides only a brief mention of the organization's strategy without defining strategic objectives	The report discusses the organization's strategy and strategic objectives but lacks clarity on how these objectives will be achieved. The strategy section may be generic and not tailored to the organization's specific circumstances. There is minimal discussion of resource allocation or how the company plans to use its <i>capitals</i> to support strategic goals	The report presents clear strategic objectives, supported by performance measures (KPIs) and a resource allocation plan linked to the <i>capitals</i> . The discussion is predominantly narrative, repetitive/ non-concise and may lack quantitative comparisons or clear timelines for achieving strategic targets	The report provides discussions of how the organization intends to achieve its strategic objectives, integrating the business model and <i>capitals</i> into the strategy execution. It includes a detailed resource allocation plan, showing how the organization prioritizes and invests in <i>capitals</i> to support value creation. The strategic objectives are linked to KPIs and are compared over time or against set targets. The strategy section demonstrates a forward-looking perspective, incorporating forecasts and resource allocation trade-offs	The report discusses how the organization plans to achieve its strategic objectives with reference to the capital and business model. Connectivity between key business processes is evident, linking strategy, risk management, financial performance and stakeholder expectations. The resource allocation plan demonstrates clear trade-offs between different <i>capitals</i> and the report provides future-oriented insights, showing how strategic objectives will be achieved across short, medium and long-term horizons. The use of graphics, data visualizations or scenario analysis further enhances stakeholder understanding of the company's strategic direction. The strategy is linked to SMART KPIs and compared against targets. Validation of strategic decision-making framework, reasonableness of financial/resource allocation projections and stress testing of long-term strategic goals

(continued)

Table A1. Continued

Grading items	1 poor	2 developing	3 designated, identifiable and informative	4 concise and insightful	5 credible, connected and intuitive
The <i>Capitals</i>	The report focuses only on financial capital, with no reference to other <i>capitals</i> (e.g. human, intellectual, social, natural, or manufactured capital)	The report acknowledges the existence of multiple <i>capitals</i> but the presentation is mostly descriptive, without clearly explaining how the <i>capitals</i> are used as inputs to the business model or how they generate value over time	The report provides a structured discussion on the relevant <i>capitals</i> as inputs to the business model and explains how they contribute to value creation. KPIs for <i>capitals</i> are included, but they may not be benchmarked against industry standards or prior performance, limiting comparability	The report presents <i>capitals</i> as both inputs and outcomes of the business model. Quantitative and qualitative KPIs are included and the KPIs are compared against industry benchmarks or internal targets, enhancing transparency, credibility and comparability. While the KPIs for relevant <i>capitals</i> connect past and present performance, scant reference is provided to expect future performance. The report also explains context and drivers behind capital changes (e.g. <i>impact of workforce restructuring on human capital</i>)	The report provides a comprehensive, connected and forward-looking discussion of both the context and drivers of changes in the <i>capitals</i> . It uses SMART KPIs to track capital performance, both quantitatively and qualitatively and explicitly links <i>capitals</i> to strategy, resource allocation and business model outcomes. The interdependencies between <i>capitals</i> are explored, demonstrating how trade-offs are managed (e.g. <i>balancing short-term financial performance with long-term environmental sustainability</i>). Time-based connectivity is evident, with capital KPIs tracked over past, present and future periods (i.e. beyond one year). Third-party validation of capital utilization, efficiency of capital trade-offs and accuracy of sustainability-related capital disclosures (e.g. carbon accounting, human capital growth)

(continued)

Table A1. Continued

Grading items	1 poor	2 developing	3 designated, identifiable and informative	4 concise and insightful	5 credible, connected and intuitive
Business Model as Part of the Value Creation Process	The report does not reference the business model or the value creation process	The report provides a basic description of the business model, focusing on what the organization does rather than how it creates value. The disclosure is largely narrative based, with no visual representation of the business model and limited discussion on key inputs and outputs	The report includes an entity-level business model diagram outlining the company's key business activities. There is some discussion of key business processes and selected KPIs are used to measure business model performance. However, the explanation of inputs (<i>capitals</i>) and outputs (value created or consumed) remains limited	The report presents the organization's business model, as part of its value creation process. It clearly articulates how <i>capitals</i> are used as inputs, how they drive critical business activities and the outputs and outcomes generated. The business model disclosure is supported by a complete entity-level diagram, with highlights on critical business activities within these processes that are material to value creation	The report discusses and connects the business model as an integrated part of value creation. The entity-level business model diagram is complemented by a high-level description of key business processes, using SMART KPIs to measure inputs, activities, outputs and outcomes. Interconnections between processes are highlighted, demonstrating how strategy, risk management, stakeholder engagement and innovation influence value creation. Verification of business model assumptions, alignment with resource allocation and validation of interdependencies between <i>capitals</i>

(continued)

Table A1. Continued

Grading items	1 poor	2 developing	3 designated, identifiable and informative	4 concise and insightful	5 credible, connected and intuitive
Risks and Opportunities	The report provides limited discussion on risks and opportunities	The report identifies major risks and opportunities but presents them in a primarily narrative format with minimal quantitative performance data. Risk mitigation strategies, if mentioned, are high-level and lack detail	The report provides a structured discussion of major risks and opportunities, including how they are identified and managed. Risk mitigation measures are discussed for key risks, and limited quantitative indicators (e.g. financial impact, probability assessments) are provided. However, there is limited analysis of how opportunities are actively pursued	The report explains how risks and opportunities are embedded in the organization's corporate governance, strategic management and materiality determination process. There is clear evidence of how risks and opportunities identified through stakeholder engagement are considered in the company's governance, strategy, business model and resource allocation decisions. The discussion includes quantitative risk metrics and provides some scenario analysis. Risks and opportunities are linked to material issues where relevant	The report utilizes both qualitative and quantitative metrics to assess risk exposure, mitigation effectiveness and opportunities. The report includes SMART KPIs for risk management and is part of highlights summary. Scenario analysis and sensitivity testing are incorporated to assess the impact of critical risks (e.g. climate change, economic downturns, regulatory shifts) on business performance. Connectivity is evident, indicating the relationships among risks as well as linking risks and opportunities to strategy, performance, governance and business model components. Additionally, risk management and opportunity maximization strategies are incorporated into executive performance targets and remuneration frameworks, demonstrating a forward-looking approach. Validation of risk identification processes, probability modelling of key risks and credibility of mitigation strategies using industry standards

(continued)

Table A1. Continued

Grading items	1 poor	2 developing	3 designated, identifiable and informative	4 concise and insightful	5 credible, connected and intuitive
Performance	The report focuses only on short-term financial performance, with no discussion of sustainability-related (medium and long-term financial) performance indicators	The report discusses performance across a range of indicators and is presented in a highlights summary, but the presentation is fragmented	The report presents a structured discussion of performance, covering both short-term financial and sustainability-related financial KPIs. The report provides comparisons with prior periods and discusses how past performance informs current strategic decisions. However, performance measurement lacks a fully balanced view and not all KPIs are explicitly linked to strategic targets, risks and opportunities, the <i>capitals</i> and key stakeholder relationships	The report covers a balanced set of KPIs across strategic objectives, <i>capitals</i> , stakeholder relationships, key business processes and risk management. Performance indicators are compared against targets, past results and, where relevant, industry benchmarks. The report explores the implications of performance trends, identifying key drivers and challenges. Forward-looking elements are included, such as projected performance goals or areas of improvement	The report demonstrates a clear link between strategic objectives, risks and opportunities, <i>capitals</i> , key stakeholder relationships and critical business activities. SMART KPIs are presented and comparable across periods and with external benchmarks. Connectivity is evident, as performance metrics link past trends, current achievements and future targets. The Highlights Summary at the front of the report balances KPIs measuring both opportunities and risk management effectiveness, ensuring an integrated perspective on performance evaluation. External verification of key performance indicators (both financial and ESG), including trend consistency and industry benchmarking

(continued)

Table A1. Continued

Grading items	1 poor	2 developing	3 designated, identifiable and informative	4 concise and insightful	5 credible, connected and intuitive
Outlook	The report provides no discussion of the organization's future outlook beyond the next financial year	While certain content elements (e.g. strategy, risks, business model) reference future plans, the report only includes some future-oriented discussion and remains qualitative and vague (e.g. <i>"We aim to enhance our product offerings over the next few years"</i>)	The report presents a structured and informative discussion of the organization's outlook, distinguishing between short-term (<12 months) and long-term (>1 year) expectations. Some forward-looking KPIs are included with clear timeframes (e.g. <i>"We are planning to develop two new products within 15 months"</i>)	The report delivers an insightful, forward-looking discussion across most content elements (e.g. strategy, risks, external environment, resource allocation, business model, performance). Future expectations are clearly defined for short (<12 months), medium (3–5 years) and long-term (>5 years) horizons. The discussion includes targeted KPIs with specific timeframes (e.g. <i>"We are planning to reengineer our product development process and develop a series of new generation products over the next 5 years and beyond, which meet customer requirement"</i>). It also considers anticipated changes over time and the key assumptions influencing the outlook. Sensitivity analysis is introduced, offering some insight into potential future scenarios	The report integrates forward-looking information across all relevant content elements. Future projections are supported by SMART KPIs across short, medium and long-term horizons. The report incorporates sensitivity analysis and scenario planning, considering the impact of external risks, technological changes, market shifts and regulatory developments. Connectivity is evident, as future changes in strategy, risks, opportunities and business model adjustments are linked together, showing how the organization adapts over time. Advanced visual tools (e.g. scenario modelling, interactive dashboards, carbon transition pathways) enhance transparency and stakeholder engagement. Where applicable, the report aligns with frameworks such as TCFD (Task Force on Climate-related Financial Disclosures) for future risk assessment. Assurance on forecasting methodologies, sensitivity analysis and forward-looking scenario modelling accuracy (e.g. climate risk assumptions, economic trend predictions)

Notes

1. The IFRS Foundation interprets sustainability as “the ability for a company to sustainably maintain resources and relationships with and manage its dependencies and impacts within its whole business ecosystem over the short, medium, and long term” (IFRS Foundation, 2022b).
2. See <https://www.ifrs.org/about-us/who-we-are/>.
3. In assessing the quality of CSR reporting, Hammond and Miles (2004) similarly identify three assessment procedures: formal monitoring protocols, CSR award schemes and academic quality assessment.
4. There were a few 3-point scores. For example, under the content element category *Organization Overview and External Environment* there was one item labelled “Mission and vision statement”, where scores were awarded as follows: 0 = no mission or vision statement; 1 = for mission or vision statement; and 2 = mission and vision statements.
5. While most scores are binary, there were exceptions where ordinal scales were used including the connectivity between content elements (0 = No connection provided; 1 = 1–2 aspects described; 2 = 3–4 aspects described; 3 = more than 4 aspects described.); and state of key stakeholder relationships (1 = Key stakeholder relationships stated; 1 = Identification of key stakeholder needs and interests provided; 1 = Organizational response to key stakeholder needs and interests provided).
6. 0 = checklist items not disclosed, 1 = general qualitative disclosure, 2 = entity-specific disclosures and 3 = detailed discussions with specific organizational actions.
7. For six items: 0 if item not reflected in the report; 1 if item appears, but the explanation is insufficient; and 2 if item is sufficiently explained. For the seventh item, “Conciseness”: 0 if number of pages is larger than 200 pages; 1 if number of pages is between 100 and 200; and 2 if page numbers is less than 100.
8. The authors state that a detailed scoring guide was prepared to guide the independent coders, “For example, in scoring the “Business Overview” component, a 0.25 score was awarded for a description of ownership structure; another 0.25 if operating structure is also disclosed; another 0.25 if principal activities and markets are disclosed; and another 0.25 if products and services are disclosed” (Zhou *et al.*, 2017, p. 108). However, the basis for scoring the remaining 30 items is not disclosed.
9. The five companies were selected because of their membership in the IIRC business network (e.g. NAB, Stockland and Slater and Gordon Lawyers) or their leadership in sustainability reporting (e.g. Westpac).
10. A few items were scored on ordinal scales, including disclosure items related to financial and social performance, which were scaled from 0 to 3 based on the time dimensions (i.e. past, current, future), and a scale of 0–6 to measure disclosure items related to environmental performance. Each of the 941 items is equally weighted, so items measured on a binary scale (0–1) received less weight in the final score than the few items rated on a wider scale.
11. The authors do not compute a total score including the four areas because “the independence and heterogeneity of the four areas, and the difference in the measurement scales would make interpretation of the results unclear” (p. 500).
12. Integrated report’s *conciseness* is measured using the natural logarithm of the number of pages in the integrated report and the Fog Index. The *completeness* of an integrated report is captured by using Bloomberg’s ESG disclosure score as a proxy for IR quality.
13. EY identifies the top 10 companies within the “Excellent” category based on three crucial areas (i.e. clear strategic focus, emphasis on value creation, and connectivity among the various elements), so the summary effectively comprises five categories.
14. It is noteworthy that only Barth *et al.* (2017) were given access to the raw EY ranking scores for their analysis. All other studies (Mans-Kemp and Van der Lugt, 2020; Maroun, 2018; Omran *et al.*, 2021; Wang *et al.*, 2020) relied on aggregate ordinal classification into one of the 5 categories to assess a firm’s adherence to the IR Framework.

15. Further discussion of each grading item is provided in the [Supplementary](#)
16. Materiality is discussed within the IR Framework under the Basis of preparation and presentation in paragraphs 4.42, 4.45 and 4.48 (IIRC, 2021)
17. An entity is identified as being an integrated reporter if its report satisfies two criteria: (1) mentions the use of the IR Framework, and (2) either includes a “value creation model” or uses “capitals”.
18. While the primary analysis focuses on the 2022 reporting period, a longitudinal extension was conducted on the same sample for 2023 and 2024. The results indicate that reporting quality scores are highly persistent over time. Specifically, the Pearson correlation for the total index score between 2022 and 2023 is $r = 0.96$, and $r = 0.86$ between 2022 and 2024. The highest stability was observed for Governance ($r = 1.00$), while the Business model as part of the value creation process showed the greatest variation ($r = 0.72$). Furthermore, two-sample t -tests confirm no significant difference in mean quality scores between 2022 and 2023 ($p = 0.52$) or 2024 ($p = 0.33$).
19. The formula for selecting the sample size is:

$$n = \frac{Z^2 \times p \times (1-p)}{E^2}$$
 where:
 n = sample size
 Z = z-score, which corresponds to the desired confidence level.
 p = the estimated proportion of the population that has the characteristic of interest.
 E = the desired margin of error.
20. As previously discussed, alternative proxies for IRQ, such as those used by [Barth et al. \(2017\)](#) and [Zhou et al. \(2017\)](#) cannot be replicated from publicly available data.
21. For the binary variable *Alt5_ReferenceIR* the Pearson correlation coefficient is undefined because the variable is constant.

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